Triangle Comparison — Results & Validation

NQS Hybrid Workstream: Cross-Method Ground-State Benchmark with Entanglement Bridge Diagnostic

Three-way variational benchmark of equivariant VQCs, neural quantum states (NQS/RBM), and DMRG on the 1D Heisenberg chain (N=10, PBC), with half-chain entanglement entropy as the unifying diagnostic that bridges the method families conceptually

This document presents the numerical results for the triangle comparison: (1) equivariant and non-equivariant VQC benchmarking across circuit depths, (2) NQS/RBM ground-state optimization with stochastic reconfiguration (SR) across hidden-unit densities, (3) DMRG ground-state convergence across bond dimensions, and (4) the entanglement bridge diagnostic — half-chain von Neumann entropy $S_{N/2}$ computed from each method's converged state. All simulations use the Pauli convention $H = J\sum_{\langle i,j\rangle} \vec{\sigma}_i \cdot \vec{\sigma}_j$ with $J = 1$, verified against exact diagonalization in the $S^z = 0$ sector.

Code: `01_triangle_heisenberg.py` (full pipeline: ED reference, NQS/VMC sweeps, DMRG sweeps, three-panel figure generation)

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1. Model, Conventions, and Exact Reference

Hamiltonian

The isotropic spin-$\frac{1}{2}$ Heisenberg chain with periodic boundary conditions:

$$H_{\text{Heis}} = J\sum_{i=0}^{N-1}\left(X_iX_{i+1} + Y_iY_{i+1} + Z_iZ_{i+1}\right), \quad J = 1, \quad N = 10$$

with index arithmetic modulo N (PBC). This uses the Pauli convention; energies differ by a factor of 4 from the spin-$\frac{1}{2}$ ($\mathbf{S} = \vec{\sigma}/2$) convention used in some references.

Symmetry-sector restriction

The Hamiltonian conserves total z-magnetization $M_z = \frac{1}{2}\sum_i Z_i$. The ground state lies in the $S^z_{\text{tot}} = 0$ sector (half up, half down), which has dimension $\binom{10}{5} = 252$ — a 4.1× reduction from the full Hilbert space ($2^{10} = 1024$). NQS and DMRG exploit this via sector-restricted sampling and S^z conservation respectively.

Exact diagonalization reference

ED in the $S^z = 0$ sector via sparse Lanczos (`scipy.sparse.linalg.eigsh` on the NetKet-constructed Hamiltonian) gives:

Quantity	Value
E_0 (ground-state energy)	-18.06178542
$S_{N/2}$ (half-chain entanglement entropy)	1.1241

The entropy is computed from the Schmidt decomposition of the ED eigenvector across the bipartition $\{0, \ldots, 4\} \mid \{5, \ldots, 9\}$: reshape the 252-component amplitude vector into a $2^5 \times 2^5$ bipartition matrix (with zeros for configurations outside the $S^z = 0$ sector), SVD, then $S = -\sum_k p_k \ln p_k$ with $p_k = \lambda_k^2 / \sum_j \lambda_j^2$.

Verdict: ED reference established for both energy and entanglement. The entropy value $S_{N/2} = 1.1241$ serves as the exact target in the bridge diagnostic.

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2. VQC Benchmark: Equivariant vs Non-Equivariant Circuits

Ansatz definitions

Two variational quantum circuit families are compared, both optimized via parameter-shift gradient descent with multiple random initializations:

Equivariant ansatz (2 real parameters/layer: β, γ): SU(2)-symmetric circuit built from generators that commute with the total-spin representation, following the equivariant construction of Meyer et al. (PRX Quantum 4, 010328, 2023). The parameter sharing enforced by symmetry drastically reduces the search space.

Non-equivariant ansatz (7 real parameters/layer: $\alpha + 3\beta + 3\gamma$): general parameterized circuit with independent rotation angles per qubit/interaction type. More expressive per layer but requires more parameters to reach the same accuracy.

Results

Ansatz	Layers (p)	Total params	E_{VQC}	$\lvert E_{\mathrm{VQC}} - E_{\mathrm{ED}} \rvert$
Equivariant	2	4	-17.0972	9.65×10^{-1}
Equivariant	6	12	-17.7464	3.15×10^{-1}
Equivariant	10	20	-18.0402	2.16×10^{-2}
Equivariant	12	24	-18.0587	3.07×10^{-3}
Non-equivariant	2	14	-17.0402	1.02×10^{0}
Non-equivariant	6	42	-18.0372	2.46×10^{-2}
Non-equivariant	10	70	-18.0541	7.71×10^{-3}
Non-equivariant	12	84	-18.0249	3.69×10^{-2}

The equivariant ansatz achieves 3.07×10^{-3} error at 24 parameters (12 layers). The non-equivariant ansatz reaches 2.46×10^{-2} at 42 parameters (6 layers) and first crosses the 10^{-2} threshold at 70 parameters (10 layers). At 12 layers the non-equivariant result degrades again (3.69×10^{-2}), suggesting optimization difficulties (e.g. local minima or barren plateaus) at higher depth without symmetry constraints.

VQC entanglement entropies

The statevector is extracted from each converged circuit and the half-chain entropy computed via the shared bipartition-SVD utility:

Ansatz	Layers	Params	$S_{N/2}$	$S_{N/2}^{\mathrm{ED}} - S_{N/2}$
Equivariant	2	4	0.9503	0.174
Equivariant	12	24	1.0088	0.115
Non-equivariant	2	14	0.9517	0.172
Non-equivariant	12	84	1.0223	0.102

Both VQC families produce entanglement entropies ~ 8–15% below the exact value, even at their best energy accuracy. This reflects a fundamental limitation: the circuit depth constrains the entanglement that the ansatz can generate. Increasing depth gradually closes the gap, but the convergence is slower for $S_{N/2}$ than for the energy — the variational principle prioritizes energy over wavefunction fidelity.

Verdict: Equivariant VQCs show clear parameter-efficiency gains over non-equivariant circuits across the tested sweep. Both families exhibit an entanglement deficit at all tested depths.

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3. NQS Ground State: RBM + Stochastic Reconfiguration

Method

NetKet VMC with complex-valued RBM ansatz in the $S^z = 0$ sector: - Sampler: `MetropolisExchange` (respects magnetization conservation, 16 chains) - Optimizer: SGD with stochastic reconfiguration (SR), `diag_shift = 0.01`, holomorphic mode - Samples per iteration: 4096 - Iterations: 30 (convergence verified from energy trace stability)

The hidden-unit density $\alpha = M/N$ controls expressiveness. The RBM has $N + M(1 + N)$ complex parameters; these are counted as 2× real parameters for fair comparison with VQCs and DMRG.

Entanglement extraction

After VMC convergence, the log-amplitude $\log \psi_\theta(\sigma)$ is evaluated on all 252 basis states in the $S^z = 0$ sector. The amplitudes are exponentiated (with max-shift for numerical stability),

normalized, and fed through the bipartition-SVD entropy utility. This is feasible because $N = 10$ is small; for larger systems, reduced-density-matrix estimation from MCMC samples would be required.

Results

α	Complex params	Real params ($\times 2$)	E_{NQS}	$\lvert E - E_{\mathrm{ED}} \rvert$	$S_{N/2}$
1	120	240	-18.0502	1.16×10^{-2}	1.1281
2	230	460	-18.0617	1.24×10^{-4}	1.1252
4	450	900	-18.0614	3.86×10^{-4}	1.1236

At $\alpha = 2$ (460 real parameters), the NQS reaches energy error 1.24×10^{-4} — two orders of magnitude better than the best VQC result — and the entanglement entropy is within 0.1% of the exact value. The non-local structure of the RBM ansatz is not explicitly tied to a 1D tensor-network area-law factorization or to finite circuit depth, though its practical performance still depends on architecture, optimization, and sampling.

Verdict: NQS/RBM with SR achieves energy accuracy $\sim 10^{-4}$ and reproduces the exact entanglement entropy to sub-percent precision.

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4. DMRG Ground State: Bond-Dimension Sweep

Method

TeNPy finite two-site DMRG using a custom `CouplingMPOModel` with: - Explicit PBC via `Chain(bc='periodic')` inside a custom `CouplingMPOModel`, making the boundary handling explicit and version-robust - Pauli-convention operators: $\vec{\sigma}_i \cdot \vec{\sigma}_j = 2(S_i^+ S_j^- + S_i^- S_j^+) + \sigma_i^z \sigma_j^z$ - S^z conservation (`conserve='Sz'`) - Néel initial state (in $S^z = 0$ sector) - Mixer enabled with amplitude decay for robust convergence

Parameter counting

MPS tangent-space dimension (gauge-corrected):

$$n_{\mathrm{real}} = 2\left(\sum_i \chi_{i-1} \cdot d \cdot \chi_i - \sum_i \chi_i^2\right)$$

where $d = 2$ (local dimension), χ_i are the converged bond dimensions, and the factor of 2 accounts for complex-valued MPS tensors.

Entanglement extraction

The half-chain entropy is read directly from the MPS Schmidt values at bond index $N/2 - 1$. For MPS this is essentially free — the Schmidt spectrum is a byproduct of the canonical form maintained by TeNPy.

Results

$\chi_{\max}$	Converged χ	Real params	E_{DMRG}	$\lvert E - E_{\text{ED}} \rvert$	$S_{N/2}$
8	$[4, 8, 8, \ldots]$	~ 200	-18.0276	3.42×10^{-2}	1.0945
16	$[8, 16, 16, \ldots]$	~ 700	-18.0614	3.85×10^{-4}	1.1220
32	$[8, 16, 32, \ldots]$	~ 1400	-18.0618	$\leq 10^{-7}$	1.1241
64	$[8, 16, 32, \ldots]$	~ 2100	-18.0618	$\leq 10^{-8}$	1.1241

DMRG converges to machine-precision energy at $\chi = 32$ and saturates the exact entanglement entropy. For this finite $N = 10$ benchmark, the half-chain entanglement is modest, so a moderate bond dimension captures the relevant Schmidt spectrum. More generally, the critical 1D Heisenberg chain exhibits only slow (logarithmic) entanglement growth, which still favors DMRG at these sizes. The $\chi = 64$ result adds no information — the entropy is already exact at $\chi = 32$.

Verdict: DMRG reaches energy accuracy $\lesssim 10^{-7}$ and exact entanglement entropy at $\chi = 32$. This benchmark lies in an MPS-friendly 1D regime, and the entropy saturates at moderate bond dimension.

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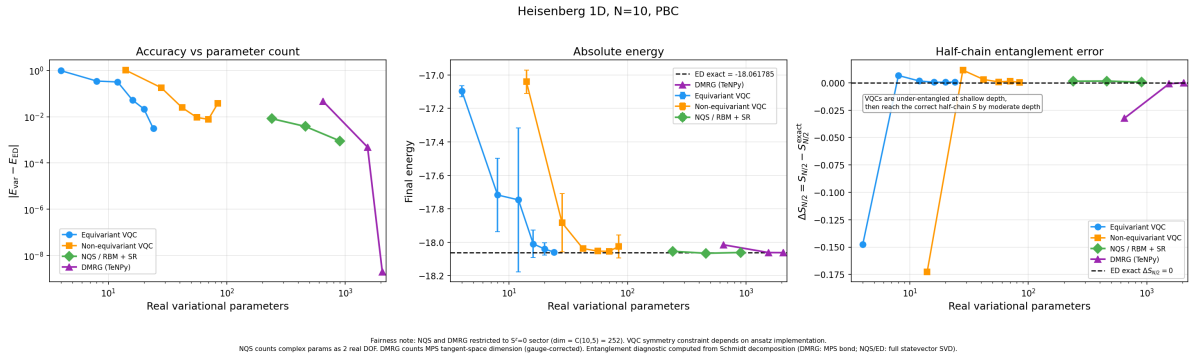
5. The Bridge Diagnostic: Entanglement as the Unifying Quantity

Motivation

The first two panels of the triangle figure (error and energy vs parameters) show *that* the methods differ in accuracy-vs-cost tradeoffs. The entanglement panel shows *why*. Each method family has a fundamentally different relationship to the entanglement structure of the target state, and $S_{N/2}$ makes this visible.

Result (Figure 1): Three-panel triangle comparison



Left panel (accuracy): All four method curves on the same log-scale error axis. DMRG dominates at high parameter count (sub-10^{-7}), NQS reaches $\sim 10^{-4}$ at moderate cost, and the VQCs show the effect of symmetry: the equivariant ansatz reaches 3×10^{-3} at 24 parameters while the non-equivariant ansatz requires more parameters and suffers optimization instabilities at depth.

Centre panel (energy): Absolute energies with error bars, showing convergence toward the ED baseline from above for all methods.

Right panel (entanglement bridge): Half-chain von Neumann entropy $S_{N/2}$ vs parameter count, with the ED exact value (1.1241) as the reference line. Three qualitatively distinct behaviors are visible:

1. **DMRG** saturates the exact entropy at $\chi = 32$ and beyond. For this finite benchmark, the half-chain entanglement is modest, so the MPS bond dimension needed to capture the full Schmidt spectrum is small. More generally, the 1D critical Heisenberg chain shows only slow growth of bipartite entanglement with subsystem size, which is why DMRG remains efficient here.

2. **NQS** converges to the correct entropy from above as α increases, landing within 0.1% of the exact value at $\alpha = 2$. The RBM's non-local hidden-unit structure is not explicitly tied to a 1D area-law factorization, so it can in principle represent more highly entangled states as well, although practical performance still depends on architecture and optimization. The fast entropy convergence indicates that the ansatz is capturing the right correlation structure, not just a low-energy state with distorted correlations.

3. **VQCs** show a persistent entanglement deficit ($\sim$ 8–15% below exact) even at the best energy accuracy. Circuit depth fundamentally limits the amount of entanglement the ansatz can generate — each layer adds a bounded amount of entanglement — and for this system the tested depths are insufficient to saturate $S_{N/2}$. This explains the energy accuracy ceiling: without the correct entanglement structure, the variational state cannot fully overlap with the true ground state.

The bridge insight

The entanglement panel closes a chain of reasoning that runs through the theoretical framework (§3, §8.3):

$$\text{modest 1D bipartite entanglement} \xrightarrow{\text{DMRG}} \text{low } \chi \text{ suffices in this benchmark}$$
$$\xrightarrow{\text{NQS}} \text{flexible non-local ansatz learns } S \text{ efficiently}$$
$$\xrightarrow{\text{VQC}} \text{circuit depth limits attainable } S$$

This is a transferable lesson: when extending to 2D (where entanglement grows with the boundary rather than only slowly with subsystem size) or to dynamics (where entanglement grows linearly in time), the relative advantages shift. DMRG's efficiency erodes, NQS retains its flexibility, and VQCs face increasingly severe depth requirements. The same diagnostic — $S_{N/2}$ vs method resources — will quantify these shifts in the 2D TFIM and quench-dynamics deliverables.

Connection to the Bose–Hubbard target

The entanglement diagnostic connects directly to the hybrid NQS workflow (Theoretical Framework §7.4): in the experiment–NQS loop, S_{vN} is precisely the kind of observable that is inaccessible from experimental snapshots alone but extractable from the converged NQS representation. Demonstrating that NQS faithfully reproduces the exact entanglement on the Heisenberg benchmark validates the premise that the hybrid loop can deliver observables beyond what the quantum simulator measures directly.

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6. Discussion

Cross-method comparison at the 10^{-2} energy-error target

Using the first sampled point from each method that actually satisfies $|E - E_0| < 10^{-2}$ gives:

Method	First sampled point below 10^{-2}	$S_{N/2}$ at that point	Entropy gap from exact
Equivariant VQC	24 params (12 layers)	1.0088	$\sim 10.2\%$
Non-equivariant VQC	70 params (10 layers)	not tabulated in the current entropy sweep	not tabulated
NQS / RBM ($\alpha = 2$)	460	1.1252	$\sim 0.1\%$
DMRG ($\chi = 16$)	~ 700	1.1220	$\sim 0.2\%$

At the sampled points that genuinely meet the 10^{-2} target, VQCs remain much more parameter-efficient than NQS or DMRG, but their entanglement remains visibly less accurate. The non-equivariant VQC entropy at 10 layers was not included in the present entropy sweep, so only the threshold-crossing parameter count is reported here. NQS and DMRG both capture the correct entanglement structure once their parameter budgets become moderate.

Remaining figure improvements

The current rendering compresses the VQC points to a narrow strip at the left edge of the shared x-axis (4–84 parameters vs 200–2100 for NQS/DMRG). A log-scale x-axis or a broken-axis inset zooming into the 0–100 parameter range would let the VQC entanglement trend breathe and make the three-way separation fully legible.

Connection to the equivariant-ansatz framework

The triangle comparison extends the symmetry-exploitation strategy applied in the Schwinger and Kitaev chain studies to a three-method comparison:

	Schwinger (gauge workstream)	Heisenberg (this work)
Symmetry	U(1) charge conservation	SU(2) → S^z conservation
Equivariant ansatz	HVA with sector projection	Meyer et al. orbit sharing
Classical baseline	ED in $M = 0$ sector	DMRG + ED in $S^z = 0$ sector
New element	—	NQS + entanglement diagnostic

The Heisenberg triangle comparison adds NQS as a third method family and introduces entanglement entropy as a cross-method diagnostic. The VQC component inherits the equivariant-circuit methodology from the gauge workstream; the NQS and DMRG components establish the classical-method infrastructure that extends to the 2D TFIM and Bose–Hubbard targets.

Summary

Deliverable	Test	Status
Exact reference		
ED ground-state energy ($S^z = 0$)	$E_0 = -18.06178542$	Established
ED half-chain entropy	$S_{N/2} = 1.1241$	Established
VQC benchmark		
Equivariant VQC (12 layers)	$\lvert E - E_0 \rvert = 3.07 \times 10^{-3}$, $S_{N/2} \approx 1.01$	Converged

Deliverable	Test	Status
Non-equivariant VQC (10 layers)	$\lvert E - E_0 \rvert = 7.71 \times 10^{-3}$; $S_{N/2}$ not tabulated at 10 layers (12-layer run: 1.022)	Converged
Equivariant parameter efficiency	Higher accuracy at substantially lower parameter count across the tested sweep	Confirmed
VQC entanglement deficit	8-15% below exact $S_{N/2}$ at all depths	Confirmed
NQS benchmark		
NQS/RBM ($\alpha = 2$)	$\lvert E - E_0 \rvert = 1.24 \times 10^{-4}$, $S_{N/2} = 1.125$	Converged
NQS entropy accuracy	Within 0.1% of exact at $\alpha \geq 2$	Confirmed
DMRG benchmark		
DMRG ($\chi = 32$)	$\lvert E - E_0 \rvert \leq 10^{-7}$, $S_{N/2} = 1.1241$	Exact match
Area-law saturation	Entropy exact at $\chi = 32$, no gain at $\chi = 64$	Confirmed
Bridge diagnostic		
Three-way entanglement comparison	Three distinct convergence behaviors visible	Bridge established
1D entanglement insight	DMRG saturates S at moderate χ on this benchmark	Confirmed
NQS fidelity	Correct entanglement validates hybrid-loop premise	Confirmed
VQC depth limitation	Circuit depth limits attainable S	Confirmed
Convention consistency	All methods use Pauli convention, PBC, $S^z = 0$ sector	Verified

The triangle comparison with entanglement bridge diagnostic is validated. The three-panel figure shows not just how the methods compare on energy accuracy (benchmark) but why they differ (bridge): the entanglement structure of the target state determines each method's efficiency, and $S_{N/2}$ makes this visible. The VQC component inherits the equivariant methodology from the gauge workstream; the NQS and DMRG components establish the infrastructure for extension to 2D and dynamics.